

Milestone 1 — A leak-free, reproducible data pipeline for MILK10k

George Hanna · Computer Vision & Speech Recognition, EADA · 4 October 2026 · reproduce with python `scripts/run_milestone1.py` (all numbers below come from its log)

Summary. All 10,480 images exist and verify; the target `diagnosis_1` keeps three classes; the data are split 70/15/15 *by lesion* with zero overlap and ≤ 0.6 pp class drift; train is imbalanced 27.6 : 1 and both a class-weighted loss and a weighted sampler are ready; images enter the model at 224×224 with train-only normalisation and label-safe augmentation.

1. Label strategy

Primary target: `diagnosis_1` with three classes. Malignant 3,634 lesions (69.4%), Benign 1,483 (28.3%), Indeterminate 123 (2.3%). All Indeterminate lesions are actinic keratoses (AKIEC, which also holds 180 malignant SCC in situ). *Dropping* them would leave the model unable to handle a lesion type it will meet in practice; *merging* them into Malignant is defensible (both lead to treatment) but hides the only pre-cancerous group, which is managed differently (cryotherapy or topical therapy rather than excision). I keep the class and handle its rarity with weights or sampling. Confusing Indeterminate with Malignant is clinically cheap (both trigger action), so a secondary binary view, "needs action" (Malignant or Indeterminate) vs. Benign, is stored for clinical reporting. **11-class stretch goal:** DF (52), INF (50) and VASC (47) are kept and up-weighted, since each has a distinct appearance. MAL_OTH (9 lesions: 6 train, 3 val, 0 test) can be neither learned nor evaluated, and its balanced weight (≈ 53) would destabilise training. It is merged with BEN_OTH (44) into **OTHER** (53 lesions, read as "unusual lesion → refer"), giving 10 classes. The trade-off: OTHER mixes benign and malignant, so it must never be read as "benign". The mapping is saved in `artifacts/label_map.json`.

2. Split design and verification

Split	Lesions	Share	Images	Malignant	Benign	Indeterminate	Rare 11-class lesions (DF / INF / VASC / BEN_OTH / MAL_OTH)
train	3,667	70.0%	7,334	69.2%	28.3%	2.5%	36 / 35 / 33 / 31 / 6
val	787	15.0%	1,574	69.8%	28.5%	1.8%	8 / 7 / 8 / 7 / 3
test	786	15.0%	1,572	69.6%	28.2%	2.2%	8 / 8 / 6 / 6 / 0

`split_lesions()` runs **StratifiedGroupKFold at lesion level** (groups = `lesion_id`, stratified on the 11 classes): it cuts 20 stratified folds of 5% each, then 3 folds become test and 3 become val (seed 42, created 2026-10-04). Images are assigned afterwards by lesion. Verified in code: zero `lesion_id` overlap between every pair of splits; every lesion has exactly 2 images in its split, one dermoscopic and one clinical; the largest deviation from the global class proportions is 0.21 pp over the 11 classes and 0.57 pp for `diagnosis_1` (Indeterminate). A property test over 10 seeds (no overlap, sizes within ± 1 pp, reproducible) passes. The files are `splits/train.csv`, `val.csv` and `test.csv`, one row per image.

3. Class imbalance on TRAIN and how it is handled

Train holds 5,076 Malignant, 2,074 Benign and 184 Indeterminate images: an **imbalance ratio of 27.6 : 1** (Malignant to Benign: 2.4 : 1). Two corrections are implemented, to be used one at a time (together they would correct twice):

- **Class-weighted loss:** weights $n / (K \cdot \text{count})$ from train only: Benign 1.18, Indeterminate 13.29, Malignant 0.48 (`artifacts/class_weights.json`).
- **WeightedRandomSampler:** in 20 train batches the label counts move from 177 / 16 / 447 (Benign / Indeterminate / Malignant, plain shuffle) to 224 / 215 / 201.

Evaluation uses balanced accuracy, macro-F1 and per-class recall per lesion, never plain accuracy (always answering "Malignant" already scores 69%).

4. Quality issues found and how they are handled

- **Files:** every row resolves to an image (0 missing) and `Image.verify()` finds 0 unreadable files; all images are 600×450 RGB (`outputs/milestone1/image_size_summary.csv`). The Session 2 loader and the new `MilkDataset` now **fail loudly** on a missing file (`FileNotFoundException`) unless `allow_missing=True` is passed explicitly.
- **Missing values:** `anatom_site_general` is 37.3% missing, but 3,850 of the 3,912 "missing" images are **trunk** lesions in `supplements/training_input.csv`: the column simply has no trunk category. Site is recovered as "trunk", with "unknown" for the other 62 images. Age (0.4% missing) is imputed with the train median if used. `anatom_site_special` (98% missing) is dropped. Missing values in `diagnosis_3/4` and `melanocytic` mean "not applicable".
- **Leakage and shortcuts:** `diagnosis_2/3/4`, `melanocytic` (True exactly for NV and MEL), `diagnosis_confirm_type` and `concomitant_biopsy` (identical columns; 87% of lesions that were not biopsied are benign) leak the label and are never inputs. `image_type` is label-free (one image of each type per lesion). `image_manipulation="altered"` is 12.8% Indeterminate vs. 2.0%, so it is not an input and is monitored as a possible shortcut. File size carries no label signal (ROC-AUC 0.48–0.51).

5. Preprocessing choices

Resolution 224×224: 100% of the images are larger (600×450), so 224 is a 2× downscale with no upsampling, and it is the native input of ImageNet-pretrained CNNs. Evaluation resizes the shorter side to 224 and crops the centre (no shape distortion). **Normalisation** uses statistics from the 7,334 *train* images only: mean [0.685, 0.527, 0.477], std [0.122, 0.131, 0.152] (`artifacts/norm_stats.json`). `eval_transform` is deterministic (asserted on two passes over the same image). **Two images per lesion:** I chose **(a)**: each image is a training sample with its lesion's label (twice the data, both views learned), and at evaluation the two views' probabilities are averaged per lesion (`aggregate_predictions`). The alternatives: **(b)** one image type only (dermoscopic is the most standardised, but this halves the data and drops the clinical context); **(c)** both views as one sample in a two-branch model (`LesionDataset`, already written): the best use of the information, but it needs both images at inference and a larger model, so it is kept for a later milestone. **Leakage risk if evaluated per image:** each lesion would count twice, giving over-confident metrics and two different answers for one patient. Had the *split* been made per image, 80% of test images would have their twin in train (Part A, A3.1).

Augmentation (train only)	Parameters	Why the label does not change
RandomResizedCrop	224, scale 0.75–1.0	keeps $\geq 75\%$ of the image: the (centred) lesion stays in view; mimics distance or zoom
Horizontal + vertical flip	$p = 0.5$ each	a lesion on the skin has no left/right or canonical "up"
RandomRotate90	0/90/180/270°	the dermoscope can be held at any angle; no interpolation, no black corners
ColorJitter	brightness 0.1, contrast 0.1, saturation 0.05, hue 0	within normal lighting variation; Part A (A3.5) showed any hue jitter ($\geq 7^\circ$) exceeds the 5° Benign/Malignant hue gap

1 original + 7 train_transform outputs (MEL, BCC and the rare VASC)

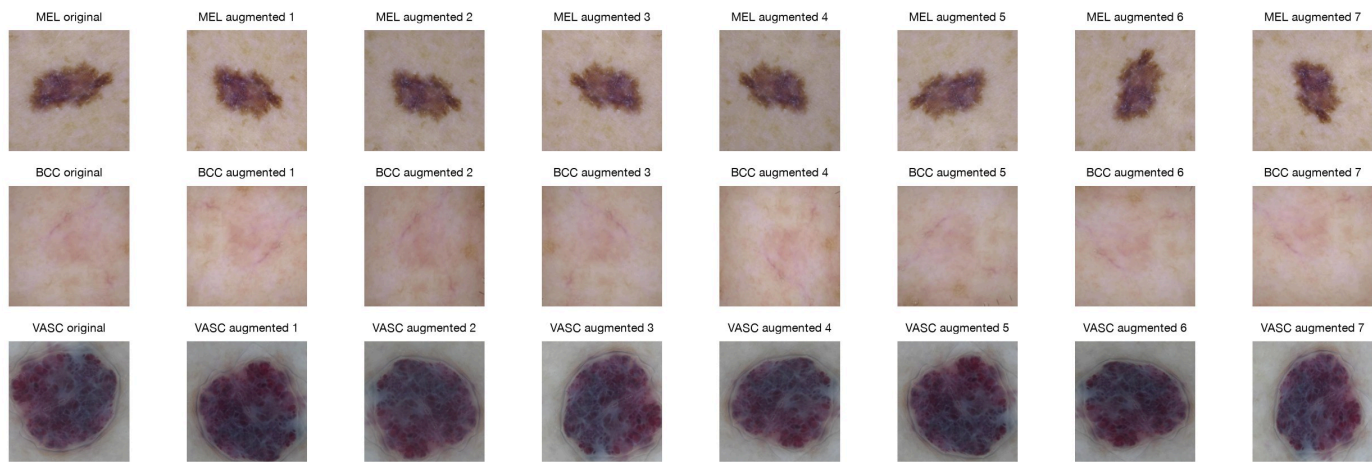


Figure 1. One original and 7 outputs of train_transform for MEL, BCC and the rare VASC: the lesion and its colours are preserved.

A train batch after train_transform (de-normalised for display)

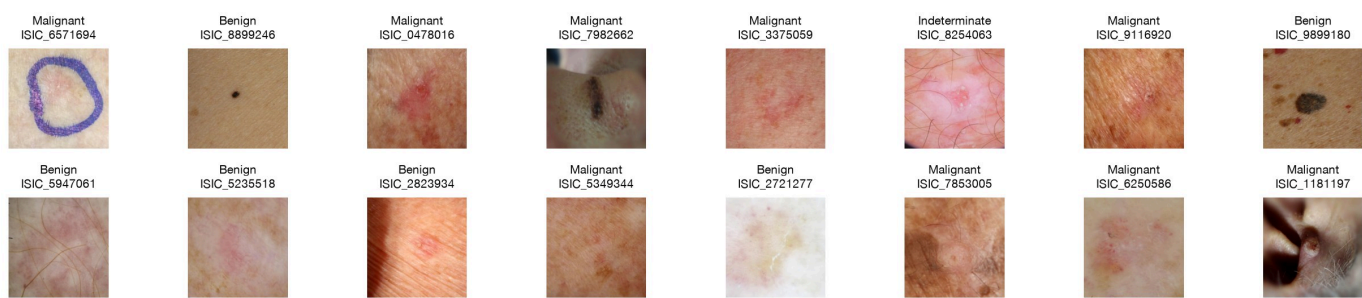


Figure 2. Visual check of a train batch from the DataLoader (de-normalised for display), drawn with the Session 2 show_grid; titles are the labels and image ids.

6. What could still go wrong

Residual leakage: MILK10k has no patient identifier, so different lesions of the same patient (same skin, camera and day) can land in different splits; near-duplicate photos across lesions have not been searched for yet (e.g. with perceptual hashes); some images show surgical pen marks around the lesion (Figure 2, first image), a known shortcut in dermatology AI that has not been audited yet; and the test split must be used once, after all tuning on validation. **Biases:** the data are biopsy-enriched (69% malignant; benign lesions are the suspicious ones), so predicted probabilities will be too high in a normal clinic and need recalibration. All images come from one study team, so devices, clinics and populations are narrow, and skin-tone coverage (skin_tone_class in the supplements) has not yet been checked. Rare classes (Indeterminate 123, INF 50, VASC 47 lesions) give noisy per-class estimates. **What a clinician should know:** this model judges lesions *that were already referred for biopsy*; it has not been validated on unselected lesions, other devices or darker skin types; it is a second opinion that may prioritise a biopsy, never a reason to skip one.